

evaluating this data, irrespective of its location. Hence, JPL has designed these networks such that the data can be streamed on a real-time basis over the Internet, and be accessible from a browser. The sensor pods are built to withstand adverse elements such as rain, temperature fluctuations, etc. JPL is looking at building the sensors using standard off-the-shelf technologies and products such as transmission and sensing equipment. So it's not going to be all that expensive to get the Sensor Web up.

Web in action

Apart from the relatively low costs, what is also very promising is the fact that the webs are up and running at several places, with some success. JPL's teams have got working Sensor Webs in place at the Huntington Gardens, California since the year 2000. Called Sensor Web 3.1, this version shows improvements in the general system design over its predecessors. JPL has designed the v3.1 pods to be relatively more impervious to the vagaries of nature and the environment in which they have been deployed, as well as making them consume less power.

The Huntington Gardens deployment covers a large area, a substantial portion of which is a public place. The pods sample and graph real-time light levels, air temperature and humidity readings. Each pod is mounted very close to the ground, and can be additionally configured to take on readings of soil temperature and soil moisture. The system also allows for monitoring of the sensor pods themselves, and researchers can run check battery status, for example, on a real-time basis. The JPL team is using the garnered

data to understand ecological balances in the area. You can dig into the Huntington project details at http://sensorwebs.jpl.nasa.gov/resources/huntington_sw31.shtml.

Sensor Webs have also been deployed at NASA's Kennedy Space Centre, Florida, located right in the midst of a large wild life refuge. Aimed at monitoring the ecology of the area, the sensor pods are mounted in the lagoons located around the shuttle launch pads. All the pods conform to the strict Federal Communications Commission (FCC) rules for radio spectrum usage, and do not interfere with the communications channels used for the space program.

It's also being designed to be deployed in very hostile territory, such as the planet Mars. To test the network over similar conditions, the JPL team set up a Sensor Web in Antarctica. With 14 self-contained battery-powered pods equipped with solar panels, temperature and humidity sensors, and onboard radio communication gear, the team aimed to observe geo-physical phenomena unique to the area. Communication between the sensors was facilitated with two dedicated communication relay pods, and one mother pod that connected to a laptop, which was used to continuously download data from the Sensor Web. The study proved the Sensor Web to be a robust platform for monitoring the physical environment, even in the frozen wastes of Antarctica.

Dust that thinks

At PARC, California, Dr. Feng Zhao and teams of scientists are working on several projects that are loosely described as Sensornets, or getting sensors to communicate and form ad-hoc networks of their own to exchange and process information. Research is now focusing on getting tiny cheap sensors—complete with battery power, networking controls and an onboard processing chip—that read, analyse and share information and yet, are so small that they could be mistaken for ordinary terrestrial dust. The applications are immense—right from animal conservation programs, to military applications,



Sensor Web pods, monitoring the environmental conditions near the Space Shuttle at the Kennedy Space Center, Florida, have been rigorously tested to ensure that they conform to all FCC regulations regarding electromagnetic interference (Sensor Webs Project, JPL, NASA)

there exists tremendous opportunity to exploit networks of information gatherers...and PARC's teams are expecting this within the next 10-odd years.

Not very different in terms of appreciable size, is Smart Dust. Researchers at the University of California's Berkeley campus are working on creating a millimetre-scale platform for a massively distributed sensor network. It will probably turn out to look like a grain of sand—maybe smaller—but Smart Dust is going to come well-wired for computing. Complete with sensors, not-insignificant computational ability and bi-directional wireless communications, this will be cheap to produce and should address a wide range of applications.

Both the PARC and Berkeley's dust motes are going to have to live with very, very small volumes. Power supplies are going to be a crunch; even with solar panels, the idea is to squeeze work out of every joule of energy. Indeed, as in the case of the Berkeley mote, at any given point of time, there will just be a clock and a few assorted timers working. All actions are initiated by a time delay, or by the motes' own initiative. Researchers are, for example, looking at the mote being smart enough to contact the base station, in case something interesting, such as a relayed report or a sensor reading, turns up.



Yet another Sensor Web pod located at the Kennedy Space Center that shows no ill effects even after 5 months of coastal weathering (Sensor Webs Project, JPL, NASA)